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Salem 12176

COMMON QUARTERLY EXAMINATION - 2026

CLASS : 12 **PHYSICS**

TIME : 3.00 Hours TOTAL: 70

I. CHOOSE THE BEST ANSWER : 15X1=15

1. An electric dipole is placed at an alignment angle of 30° with an electric field of $2 \times 10^5 NC^{-1}$. It experiences torque equal to 8 Nm. The charge on the dipole if the dipole length is 1 cm is
(a) 4 mC (b)7 mC (c)5 mC (d) 8 mC
2. The capacitance of a capacitor is
a) directly proportional to the charge q given to it b) inversely proportional to its potential V
c) directly proportional to the charge q and inversely proportional to the potential V
d) independent of both the charge q and potential V
3. Two metallic spheres of radii 1 cm and 3 cm are given charges of -1×10^{-2} C and 5×10^{-2} C respectively. If these are connected by a conducting wire, the final charge on the bigger sphere is
(a) 3×10^{-2} C (b) 4×10^{-2} C (c) 1×10^{-2} C (d) 2×10^{-2} C
4. A carbon resistor of (47 ± 4.7)k Ωto be marked with rings of different colours for its identification. The colour code sequence will be
a) Yellow – Green –Violet - Gold b) Yellow - Violet –Orange - Silver
c) Violet - Yellow - Orange – Silver d) Green - Orange – Violet - Gold
5. The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10Ω is
a) 0.2 Ω b) 0.5 Ω c) 0.8Ω d) 1.0Ω
6. A non-conducting charged ring carrying a charge of q, mass m and radius r is rotated about it's axis with constant angular speed ω. Find the ratio of its magnetic moment with angular momentum is
(a) $\frac{q}{m}$ (b) $\frac{2q}{m}$ (c) $\frac{q}{2m}$ (d) $\frac{q}{4m}$
7. The vertical component of Earths magnetic field at a place is equal to the horizontal component. What is the value of angle of dip at this place?
(a) 30^0 (b) 45^0 (c) 60^0 (d) 90^0
8. An electron is moving with a velocity of 3×10^6 ms^{-1} perpendicular to a uniform magnetic field of induction 0.5 T. The force experienced by the electron is
a) 2.4×10^{-13} N b) 13.6×10^{-27} N c) 13.6×10^{-11} N d) zero
9. In a transformer, the number of turns in the primary and the secondary are 410 and 1230 respectively. If the current in primary is 6A, then that in the secondary coil is
(a) 2 A (b) 18 A (c) 12 A (d) 1 A
10. In an oscillating LC circuit, the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is
(a) $\frac{Q}{2}$ (b) $\frac{Q}{\sqrt{3}}$ (c)Q (d) $\frac{Q}{\sqrt{2}}$
11. The instantaneous emf and current equations of an a.c. circuit are respectively e = 200 sin $\left(\omega t + \frac{\pi}{3}\right)$ and i = 20 sin ωt. The average power consumed over one complete cycle is
a) 2000 W b) 1000 W c) 4000 W d) 707 W
12. Which of the following electromagnetic radiation is used for viewing objects through fog
(a) microwave (b)infrared (c) X- rays (d) gamma rays
13. In an electromagnetic wave the phase difference between electric field $\vec{E}$ and magnetic field $\vec{B}$ is
a) π/4 b) π/2 c)π d) zero
14. For light incident from air onto a slab of refractive index 2. Maximum possible angle of refraction is,
(a) 30^0 (b) 45^0 (c) 60^0 (d) 90^0

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15. A ray of light passes form a denser medium into a rarer medium. For an angle of incidence of 45° the refracted ray grazes the surface of separation of the two media. The refractive index of the denser medium is

a) 3/2 b) $1/\sqrt{2}$ c) $\sqrt{2}$ d) 2

ANSWER ANY SIX QUESTIONS. QUESTION No. 24 COMPULSORY: 6X2=12

16. Define 'capacitance'.Giveits unit.
17. What is corona discharge?
18. Define mobility. Give its unit.
19. Calculate the equivalent resistance for the circuit which is connected to 24V battery and also find the potential difference across 4Ω and 6Ω resistors in the circuit.
20. State Ampere's circuital law.
21. Mention the ways of producing induced emf.
22. Why are Electromagnetic waves non-mechanical?
23. Why does sky appear blue?
24. For a transparent medium relative permeability and permittivity μ_r and ε_r are 1.0 and 1.44 respectively, then what is the velocity of the light in a medium ? .

ANSWER ANY SIX QUESTIONS. QUESTION No. 33 COMPULSORY: 6X3=18

25. Obtain Gauss's law of electrostatics from Coulomb's law.
26. A parallel plate capacitor has square plates of side 5 cm and separated by a distance of 1 mm. (a) Calculate the capacitance of this capacitor. (b) If a 10 V battery is connected to the capacitor, what is the charge stored in any one of the plates?
27. Obtain the Relation between current and drift velocity.
28. Discuss the conversion of galvano meter in to a voltmeter.
29. Explain the various energy losses in a transformer.
30. Find out the phase relationship between voltage and current in a pure resistor circuit.
31. What are Fraunhofer lines? How are they useful in the identification of elements present in the Sun?
32. Derive the relation between f and R for a spherical mirror.
33. The repulsive force between two magnetic poles in air is 9×10^{-3}N. If the two poles are equal in strength hand are separated by a distance of 10cm, calculate the pole strength of each pole.

ANSWER ALL THE QUESTIONS: 5X5=25

34. Derive an expression for electrostatic potential due to an electric dipole. (OR)
Explain in detail the effect of introducing a dielectric medium between the plates of a parallel plate capacitor, when the capacitor is disconnected from the battery.
35. How the emf of two cells are compared using potentiometer? (OR)
Discuss the working of cyclotron in detail.
36. Calculate the magnetic field at a point on the axial line of a bar magnet.
(OR)
Derive an expression for phase angle between the applied voltage and current in a series RLC circuit.
37. What is absorption spectrum? Explains its types.
(OR)
What is dispersion? Obtain the equation for dispersive power of a medium.
38. (i)Why nichrome wire usd as heating element.
(ii)The resistance of a nichrome wire at 20°C is 20 Ω.If its temperature coefficient of resistivity of nichrome is 0.005/°C, find the resistance of the wire at boiling point of water. Comment on the result.
(OR)
(i) Define average value of alternating current.
(ii)The equation for an alternating current is given by $i = 66\sin 314t$. Find the peak current, frequency, and instantaneous value of current at t = 1 ms.

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